

Generative Adversarial Networks

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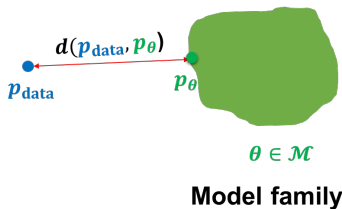
Lecture 10

- <https://github.com/hindupuravinash/the-gan-zoo>
The GAN Zoo: List of all named GANs
- Today
 - Rich class of likelihood-free objectives via f -GANs
 - Inferring latent representations via BiGAN
 - Application: Image-to-image translation via CycleGANs

Beyond KL and Jensen-Shannon Divergence



$$\mathbf{x}^{(j)} \sim p_{\text{data}} \\ j = 1, 2, \dots, |\mathcal{D}|$$



What choices do we have for $d(\cdot)$?

- KL divergence: Autoregressive Models, Flow models
- (scaled and shifted) Jensen-Shannon divergence: original GAN objective

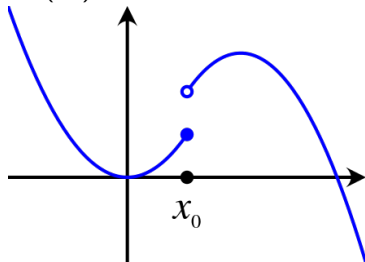
f divergences

- Given two densities p and q , the f -divergence is given by

$$D_f(p, q) = E_{\mathbf{x} \sim q} \left[f \left(\frac{p(\mathbf{x})}{q(\mathbf{x})} \right) \right]$$

where f is any convex, lower-semicontinuous function with $f(1) = 0$.

- Convex: Line joining any two points lies above the function
- Lower-semicontinuous: function value at any point $\mathbf{x}_0$ is close to $f(\mathbf{x}_0)$ or greater than $f(\mathbf{x}_0)$



- Example: KL divergence with $f(u) = u \log u$

f divergences

Many more f-divergences!

Name	$D_f(P\ Q)$	Generator $f(u)$
Total variation	$\frac{1}{2} \int p(x) - q(x) \, dx$	$\frac{1}{2} u - 1 $
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} \, dx$	$u \log u$
Reverse Kullback-Leibler	$\int q(x) \log \frac{q(x)}{p(x)} \, dx$	$-\log u$
Pearson χ^2	$\int \frac{(q(x) - p(x))^2}{p(x)} \, dx$	$(u - 1)^2$
Neyman χ^2	$\int \frac{(p(x) - q(x))^2}{q(x)} \, dx$	$\frac{(1-u)^2}{u}$
Squared Hellinger	$\int \left(\sqrt{p(x)} - \sqrt{q(x)} \right)^2 \, dx$	$(\sqrt{u} - 1)^2$
Jeffrey	$\int (p(x) - q(x)) \log \left(\frac{p(x)}{q(x)} \right) \, dx$	$(u - 1) \log u$
Jensen-Shannon	$\frac{1}{2} \int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \, dx$	$-(u + 1) \log \frac{1+u}{2} + u \log u$
Jensen-Shannon-weighted	$\int p(x) \pi \log \frac{p(x)}{\pi p(x) + (1-\pi)q(x)} + (1-\pi)q(x) \log \frac{q(x)}{\pi p(x) + (1-\pi)q(x)} \, dx$	$\pi u \log u - (1-\pi + \pi u) \log(1-\pi + \pi u)$
GAN	$\int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \, dx - \log(4)$	$u \log u - (u + 1) \log(u + 1)$
α -divergence ($\alpha \notin \{0, 1\}$)	$\frac{1}{\alpha(\alpha-1)} \int \left(p(x) \left[\left(\frac{q(x)}{p(x)} \right)^\alpha - 1 \right] - \alpha(q(x) - p(x)) \right) \, dx$	$\frac{1}{\alpha(\alpha-1)} (u^\alpha - 1 - \alpha(u - 1))$

Source: Nowozin et al., 2016

f -GAN: Variational Divergence Minimization

- To use f -divergences as a two-sample test objective for likelihood-free learning, we need to be able to estimate it only via samples
- Fenchel conjugate: For any function $f(\cdot)$, its convex conjugate is defined as

$$f^*(t) = \sup_{u \in \text{dom}_f} (ut - f(u))$$

- Duality: $f^{**} = f$. When $f(\cdot)$ is convex, lower semicontinuous, so is $f^*(\cdot)$

$$f(u) = \sup_{t \in \text{dom}_{f^*}} (tu - f^*(t))$$

f -GAN: Variational Divergence Minimization

- We can obtain a lower bound to any f -divergence via its Fenchel conjugate

$$\begin{aligned} D_f(p, q) &= E_{\mathbf{x} \sim q} \left[f \left(\frac{p(\mathbf{x})}{q(\mathbf{x})} \right) \right] \\ &= E_{\mathbf{x} \sim q} \left[\sup_{t \in \text{dom}_{f^*}} \left(t \frac{p(\mathbf{x})}{q(\mathbf{x})} - f^*(t) \right) \right] \\ &:= E_{\mathbf{x} \sim q} \left[T(\mathbf{x}) \frac{p(\mathbf{x})}{q(\mathbf{x})} - f^*(T(\mathbf{x})) \right] \\ &= \int_{\mathcal{X}} [T(\mathbf{x})p(\mathbf{x}) - f^*(T(\mathbf{x}))q(\mathbf{x})] d\mathbf{x} \\ &\geq \sup_{T \in \mathcal{T}} \int_{\mathcal{X}} (T(\mathbf{x})p(\mathbf{x}) - f^*(T(\mathbf{x}))q(\mathbf{x})) d\mathbf{x} \\ &= \sup_{T \in \mathcal{T}} (E_{\mathbf{x} \sim p} [T(\mathbf{x})] - E_{\mathbf{x} \sim q} [f^*(T(\mathbf{x}))]) \end{aligned}$$

where $\mathcal{T} : \mathcal{X} \mapsto \mathbb{R}$ is an arbitrary class of functions

- **Note:** Lower bound is likelihood-free w.r.t. p and q

f -GAN: Variational Divergence Minimization

- Variational lower bound

$$D_f(p, q) \geq \sup_{T \in \mathcal{T}} (E_{\mathbf{x} \sim p} [T(\mathbf{x})] - E_{\mathbf{x} \sim q} [f^*(T(\mathbf{x}))])$$

- Choose any f -divergence
- Let $p = p_{\text{data}}$ and $q = p_G$
- Parameterize T by ϕ and G by θ
- Consider the following f -GAN objective

$$\min_{\theta} \max_{\phi} F(\theta, \phi) = E_{\mathbf{x} \sim p_{\text{data}}} [T_{\phi}(\mathbf{x})] - E_{\mathbf{x} \sim p_{G_{\theta}}} [f^*(T_{\phi}(\mathbf{x}))]$$

- Generator G_{θ} tries to minimize the divergence estimate and discriminator T_{ϕ} tries to tighten the lower bound

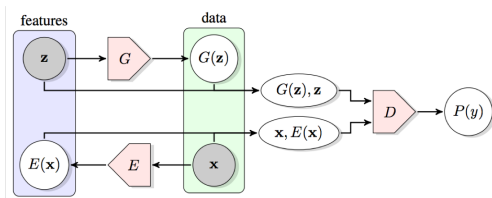
Inferring latent representations in GANs

- The generator of a GAN is typically a directed, latent variable model with latent variables $\mathbf{z}$ and observed variables $\mathbf{x}$. How can we infer the latent feature representations in a GAN?
- Unlike a normalizing flow model, the mapping $G : \mathbf{z} \mapsto \mathbf{x}$ need not be invertible
- Unlike a variational autoencoder, there is no inference network $q(\cdot)$ which can learn a variational posterior over latent variables
- **Solution 1:** For any point $\mathbf{x}$, use the activations of the prefinal layer of a discriminator as a feature representation
- Intuition: Similar to supervised deep neural networks, the discriminator would have learned useful representations for $\mathbf{x}$ while distinguishing real and fake $\mathbf{x}$

Inferring latent representations in GANs

- If we want to directly infer the latent variables $\mathbf{z}$ of the generator, we need a different learning algorithm
- A regular GAN optimizes a two-sample test objective that compares samples of $\mathbf{x}$ from the generator and the data distribution
- **Solution 2:** To infer latent representations, we will compare samples of $\mathbf{x}, \mathbf{z}$ from the joint distributions of observed and latent variables as per the model and the data distribution
- For any $\mathbf{x}$ generated via the model, we have access to $\mathbf{z}$ (sampled from a simple prior $p(\mathbf{z})$)
- For any $\mathbf{x}$ from the data distribution, the $\mathbf{z}$ is however unobserved (latent)

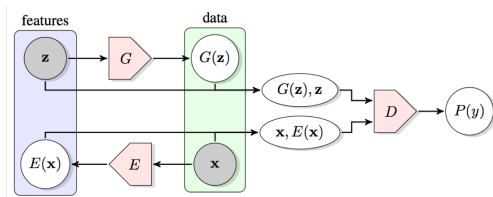
Bidirectional Generative Adversarial Networks (BiGAN)



Source: Donahue et al., 2016

- In a BiGAN, we have an encoder network E in addition to the generator network G
- The encoder network only observes $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ during training to learn a mapping $E : \mathbf{x} \mapsto \mathbf{z}$
- As before, the generator network only observes the samples from the prior $\mathbf{z} \sim p(\mathbf{z})$ during training to learn a mapping $G : \mathbf{z} \mapsto \mathbf{x}$

Bidirectional Generative Adversarial Networks (BiGAN)

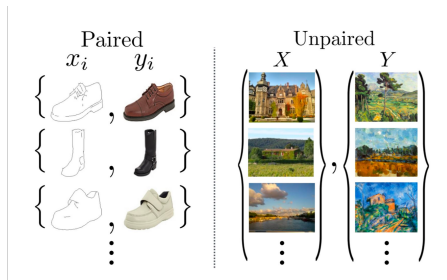


Source: Donahue et al., 2016

- The discriminator D observes samples from the generative model $z, G(z)$ and the encoding distribution $E(x), x$
- The goal of the discriminator is to maximize the two-sample test objective between $z, G(z)$ and $E(x), x$
- After training is complete, new samples are generated via G and latent representations are inferred via E

Translating across domains

- Image-to-image translation: We are given images from two domains, $\mathcal{X}$ and $\mathcal{Y}$
- Paired vs. unpaired examples

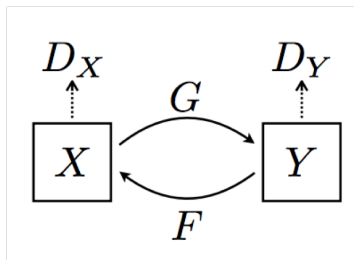


Source: Zhu et al., 2016

- Paired examples can be expensive to obtain. Can we translate from $\mathcal{X} \leftrightarrow \mathcal{Y}$ in an unsupervised manner?

CycleGAN: Adversarial training across two domains

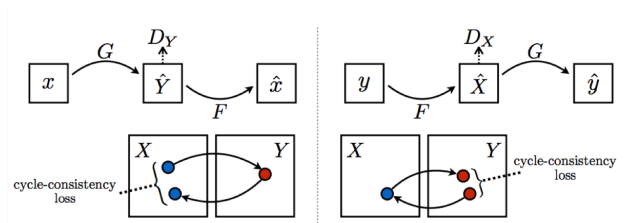
- To match the two distributions, we learn two parameterized conditional generative models $G : \mathcal{X} \leftrightarrow \mathcal{Y}$ and $F : \mathcal{Y} \leftrightarrow \mathcal{X}$
- G maps an element of $\mathcal{X}$ to an element of $\mathcal{Y}$. A discriminator D_Y compares the observed dataset Y and the generated samples $\hat{Y} = G(X)$
- Similarly, F maps an element of $\mathcal{Y}$ to an element of $\mathcal{X}$. A discriminator D_X compares the observed dataset X and the generated samples $\hat{X} = F(Y)$



Source: Zhu et al., 2016

CycleGAN: Cycle consistency across domains

- **Cycle consistency:** If we can go from X to $\hat{Y}$ via G , then it should also be possible to go from $\hat{Y}$ back to X via F
 - $F(G(X)) \approx X$
 - Similarly, vice versa: $G(F(Y)) \approx Y$

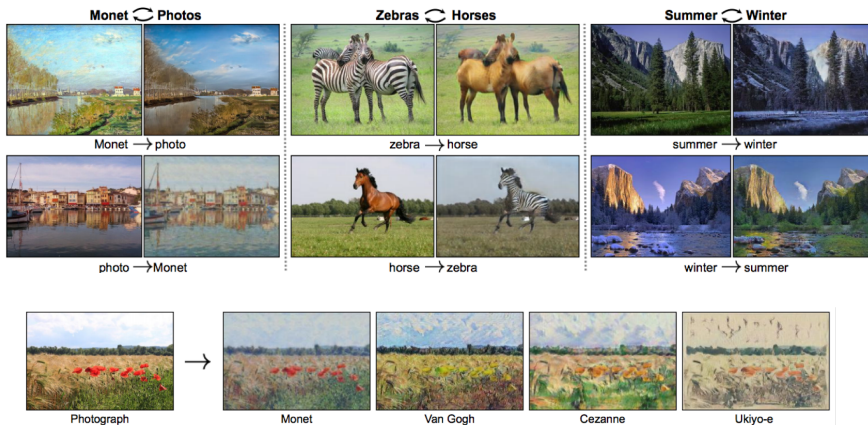


Source: Zhu et al., 2016

- Overall loss function

$$\min_{F, G, D_X, D_Y} \mathcal{L}_{\text{GAN}}(G, D_Y, X, Y) + \mathcal{L}_{\text{GAN}}(F, D_X, X, Y) \\ + \underbrace{\lambda (E_X[\|F(G(X)) - X\|_1] + E_Y[\|G(F(Y)) - Y\|_1])}_{\text{cycle consistency}}$$

CycleGAN in practice



Source: Zhu et al., 2016

Summary of Generative Adversarial Networks

- Key observation: Samples and likelihoods are not correlated in practice
- Two-sample test objectives allow for learning generative models only via samples (likelihood-free)
- Wide range of two-sample test objectives covering f -divergences (and more)
- Latent representations can be inferred via BiGAN
- Interesting applications such as CycleGAN